

Logistic Regression-Full Worked Example

Prepared by: Nafiz Imtiaz Rafin

Problem setup

We use a logistic (binary) regression model with bias and two features:

$$\hat{y} = \sigma(z) = \frac{1}{1 + e^{-z}}, \quad z = b_0 + w_1x_1 + w_2x_2.$$

Loss (binary cross-entropy / negative log-likelihood) over n samples:

$$L(b_0, w_1, w_2) = - \sum_{i=1}^n [y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)].$$

For convenience we will also use the common gradient identity for logistic loss:

$$\frac{\partial L}{\partial z_i} = \hat{y}_i - y_i,$$

and hence for parameters:

$$\frac{\partial L}{\partial b_0} = \sum_{i=1}^n (\hat{y}_i - y_i), \quad \frac{\partial L}{\partial w_j} = \sum_{i=1}^n (\hat{y}_i - y_i) x_j^{(i)}.$$

Gradient descent update (learning rate α):

$$\theta \leftarrow \theta - \alpha \nabla_{\theta} L, \quad \theta = [b_0, w_1, w_2]^{\top}.$$

Dataset (4 samples, 2 features)

i	x_1	x_2	y
1	1	2	1
2	2	0	1
3	3	1	0
4	0	2	0

Initial parameters and learning rate:

$$b_0^{(0)} = 0, \quad w_1^{(0)} = 0, \quad w_2^{(0)} = 0, \quad \alpha = 0.1.$$

Step A — Compute initial predictions and loss

At initialization $z_i^{(0)} = 0$ for all i , because all parameters are zero. Therefore

$$\hat{y}_i^{(0)} = \sigma(0) = \frac{1}{1 + e^0} = \frac{1}{2} = 0.5.$$

Compute the per-sample log-loss (binary cross-entropy) at $p = 0.5$:

$$\ell_i^{(0)} = -[y_i \log(\hat{y}_i^{(0)}) + (1 - y_i) \log(1 - \hat{y}_i^{(0)})].$$

Since $\log(0.5) = -\ln 2 \approx -0.69314718056$, each sample loss is

$$\ell_i^{(0)} = -[y_i(-\ln 2) + (1 - y_i)(-\ln 2)] = -(-\ln 2) = \ln 2 \approx 0.69314718056.$$

Thus total loss (sum over 4 samples):

$$L^{(0)} = 4 \times \ln 2 \approx 4 \times 0.69314718056 = 2.77258872224.$$

(We will keep a few decimal places when showing numeric steps below.)

Step B — Compute gradients at initialization

We compute $\hat{y}_i^{(0)} - y_i$ for each sample:

$$\begin{aligned}\hat{y}_1^{(0)} - y_1 &= 0.5 - 1 = -0.5, \\ \hat{y}_2^{(0)} - y_2 &= 0.5 - 1 = -0.5, \\ \hat{y}_3^{(0)} - y_3 &= 0.5 - 0 = +0.5, \\ \hat{y}_4^{(0)} - y_4 &= 0.5 - 0 = +0.5.\end{aligned}$$

Now sum to get gradient components.

Gradient w.r.t. b_0 :

$$\left. \frac{\partial L}{\partial b_0} \right|_{(0)} = \sum_{i=1}^4 (\hat{y}_i^{(0)} - y_i) = (-0.5) + (-0.5) + 0.5 + 0.5 = 0.$$

Gradient w.r.t. w_1 :

$$\begin{aligned}\left. \frac{\partial L}{\partial w_1} \right|_{(0)} &= \sum_{i=1}^4 (\hat{y}_i^{(0)} - y_i) x_1^{(i)} \\ &= (-0.5) \cdot 1 + (-0.5) \cdot 2 + (0.5) \cdot 3 + (0.5) \cdot 0 \\ &= -0.5 - 1.0 + 1.5 + 0 = 0.\end{aligned}$$

Gradient w.r.t. w_2 :

$$\begin{aligned}\left.\frac{\partial L}{\partial w_2}\right|_{(0)} &= \sum_{i=1}^4 (\hat{y}_i^{(0)} - y_i) x_2^{(i)} \\ &= (-0.5) \cdot 2 + (-0.5) \cdot 0 + (0.5) \cdot 1 + (0.5) \cdot 2 \\ &= -1.0 + 0 + 0.5 + 1.0 = 0.5.\end{aligned}$$

Collecting gradients at iteration 0:

$$\nabla L^{(0)} = \begin{bmatrix} \frac{\partial L}{\partial b_0} \\ \frac{\partial L}{\partial w_1} \\ \frac{\partial L}{\partial w_2} \end{bmatrix}_{(0)} = \begin{bmatrix} 0 \\ 0 \\ 0.5 \end{bmatrix}.$$

Step C — One gradient descent update

Apply the update rule:

$$\theta^{(1)} = \theta^{(0)} - \alpha \nabla L^{(0)}.$$

With $\alpha = 0.1$ and $\theta^{(0)} = [0, 0, 0]$, we get:

$$\begin{aligned}b_0^{(1)} &= 0 - 0.1 \times 0 = 0, \\ w_1^{(1)} &= 0 - 0.1 \times 0 = 0, \\ w_2^{(1)} &= 0 - 0.1 \times 0.5 = -0.05.\end{aligned}$$

So after one update:

$$\boxed{b_0^{(1)} = 0, \quad w_1^{(1)} = 0, \quad w_2^{(1)} = -0.05}.$$

Step D — Predictions and loss after the update

Compute the linear score $z_i^{(1)}$ and probability $\hat{y}_i^{(1)} = \sigma(z_i^{(1)})$ for each sample.

Sample 1: $(x_1, x_2, y) = (1, 2, 1)$

$$z_1^{(1)} = b_0^{(1)} + w_1^{(1)} \cdot 1 + w_2^{(1)} \cdot 2 = 0 + 0 \cdot 1 + (-0.05) \cdot 2 = -0.10,$$

$$\hat{y}_1^{(1)} = \sigma(-0.10) = \frac{1}{1 + e^{0.10}} \approx 0.4750208130.$$

Sample 2: $(2, 0, 1)$

$$z_2^{(1)} = 0 + 0 \cdot 2 + (-0.05) \cdot 0 = 0,$$

$$\hat{y}_2^{(1)} = \sigma(0) = 0.5.$$

Sample 3: $(3, 1, 0)$

$$z_3^{(1)} = 0 + 0 \cdot 3 + (-0.05) \cdot 1 = -0.05,$$

$$\hat{y}_3^{(1)} = \sigma(-0.05) = \frac{1}{1 + e^{0.05}} \approx 0.4875026030.$$

Sample 4: $(0, 2, 0)$

$$z_4^{(1)} = 0 + 0 \cdot 0 + (-0.05) \cdot 2 = -0.10,$$

$$\hat{y}_4^{(1)} = \sigma(-0.10) \approx 0.4750208130.$$

(Values shown to 10 decimal places; intermediate exponentials used implicitly.)

Now compute per-sample log-loss after the update:

$$\ell_i^{(1)} = -[y_i \log(\hat{y}_i^{(1)}) + (1 - y_i) \log(1 - \hat{y}_i^{(1)})].$$

Compute numerically (rounded to 6 decimals for readability):

$$\ell_1^{(1)} = -\log(\hat{y}_1^{(1)}) = -\log(0.4750208130) \approx 0.744397,$$

$$\ell_2^{(1)} = -\log(0.5) \approx 0.693147,$$

$$\ell_3^{(1)} = -\log(1 - \hat{y}_3^{(1)}) = -\log(1 - 0.4875026030) = -\log(0.5124973970) \approx 0.668103,$$

$$\ell_4^{(1)} = -\log(1 - \hat{y}_4^{(1)}) = -\log(1 - 0.4750208130) = -\log(0.5249791870) \approx 0.644394.$$

Sum to obtain total loss after one update:

$$\begin{aligned} L^{(1)} &= \ell_1^{(1)} + \ell_2^{(1)} + \ell_3^{(1)} + \ell_4^{(1)} \\ &\approx 0.744397 + 0.693147 + 0.668103 + 0.644394 \\ &\approx 2.750041. \end{aligned}$$

Step E — Summary (iteration $0 \rightarrow 1$)

Initial parameters: $b_0^{(0)} = 0$, $w_1^{(0)} = 0$, $w_2^{(0)} = 0$,

Initial loss: $L^{(0)} \approx 2.772589$,

Gradient at iteration 0: $\nabla L^{(0)} = [0, 0, 0.5]^\top$,

Learning rate: $\alpha = 0.1$,

Parameters after 1 GD update: $b_0^{(1)} = 0$, $w_1^{(1)} = 0$, $w_2^{(1)} = -0.05$,

Loss after 1 update: $L^{(1)} \approx 2.750041$.

Interpretation

- With parameters initialized at zero, every prediction was 0.5 and the initial loss was $4 \ln 2 \approx 2.7726$.
- The gradient at initialization pointed only in the w_2 direction (value 0.5) because feature x_2 differed systematically between classes in this dataset.
- After one gradient-descent step with $\alpha = 0.1$, w_2 decreased to -0.05 , slightly lowering the predicted probabilities for samples with larger x_2 values (which in our dataset correspond to class 0 examples). This yielded a small reduction in total log-loss from ≈ 2.7726 to ≈ 2.7500 .

- The example shows the mechanics: compute $\hat{y}$, compute $(\hat{y} - y)$, sum weighted by features for gradients, update parameters, recompute probabilities and loss.